



ISO/TC 12 "Quantities and units"

Secretariat: SIS

Committee manager: Radhe Sven Mr



JCGM GUM Liaison report Plenary meeting May 2025 (LP)

Document type	Related content	Document date	Expected action
Meeting / Liaison report	Meeting: VIRTUAL 22 May 2025	2025-05-09	INFO by 2025-05-22

Description

Liaison reports to
ISO/TC12 Quantities & Units
JCGM GUM & JCGM VIM
ISO/IEC JTC1 SC32: "Data management and interchange"
ISO/TC 37, Language and terminology
ISO/TC 215, Health informatics

L R Pendrill
ISO TC12 past Chair



Liaison reports to **ISO/TC12 Quantities & Units**

JCGM GUM & JCGM VIM

ISO/IEC JTC1 SC32: "Data management and interchange"

ISO/TC 37, Language and terminology

ISO/TC 215, Health informatics

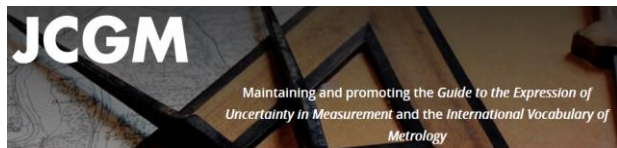
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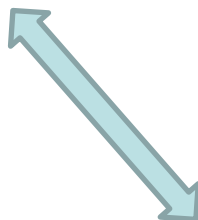
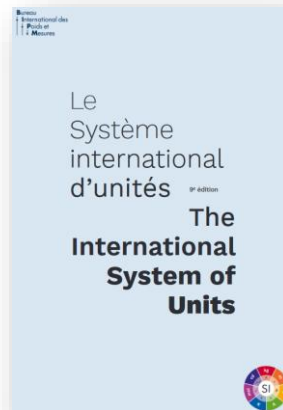


Quantities & units

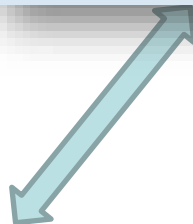
Guide to expression of uncertainty in measurement



**JCGM Working Group on the
Expression of Uncertainty in
Measurement (JCGM-WG1:GUM)**



ISO/TC 12
Quantities and units



GUM IN THE NEW PERSPECTIVE

Part 1 Introduction (revised edition of JCGM 104)

Part 2 Concepts (ex JCGM 105)

Part 3 Current 'grandfathered' GUM (ex JCGM 100)

Part 4 Role of measurement uncertainty in conformity assessment (ex JCGM 106)

Part 5 Examples of uncertainty evaluation (ex JCGM 110)

Part 6 Developing and using measurement models (ex JCGM 103)

Part 7 Propagation of distributions using a Monte Carlo method (ex JCGM 101)

Part 8 Extension to any number of output quantities (ex JCGM 102)

Part 9 Statistical models and data analysis for interlaboratory studies (ex JCGM 109)

Part 10 Least squares methods (ex JCGM 107)

Part 11 Bayesian methods (ex JCGM 108)

Part 12 Basic method for uncertainty propagation

Template for comments and secretariat observations

Date: 14/9/2023

Document: **JCGM-WG2:VIM-2023-01**

Project: International
Vocabulary of Metrology

MB/ NC ¹	Line number	Clause/ Subclause	Paragraph/ Figure/ Table/	Type of comment ²	Comments	Proposed change	Observations of the secretariat
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UPCOMING MEETING

JCGM Webinar on “Measurement Uncertainty”

02
JULY
2025

- The VIM4 3CD will be circulated for comments in the third quarter of 2024 following efforts to build consensus between the WG1 and WG2 communities on “measurement uncertainty” and associated definitions.



ROBERT SITTON



JCGM-WEBINAR



TC12
Quantities & units

ISO/IEC JTC1

ISO/IEC JTC1 SC32: "Data management and interchange"

Liaison report ISO/IEC JTC1 to ISO/TC12 Quantities & Units

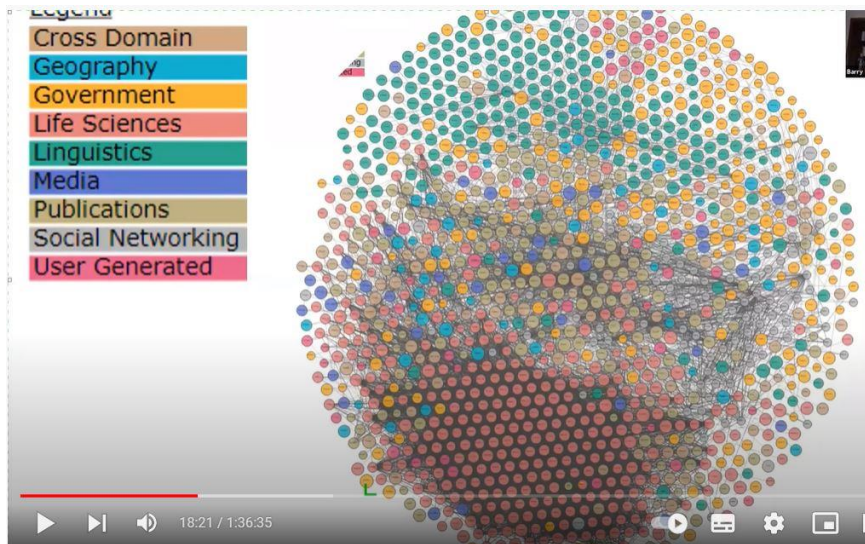
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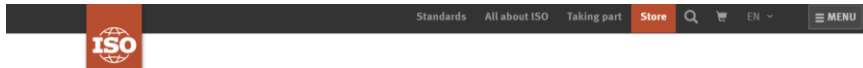
ISO/IEC DIS 21838-2

INFORMATION TECHNOLOGY -- TOP-LEVEL ONTOLOGIES
(TLO) -- PART 2: BASIC FORMAL ONTOLOGY (BFO)

iso.org/standard/74572.html



top-level ontology (TLO) =def. *ontology* that is created to represent the *categories* that are shared across a maximally broad range of *domains*

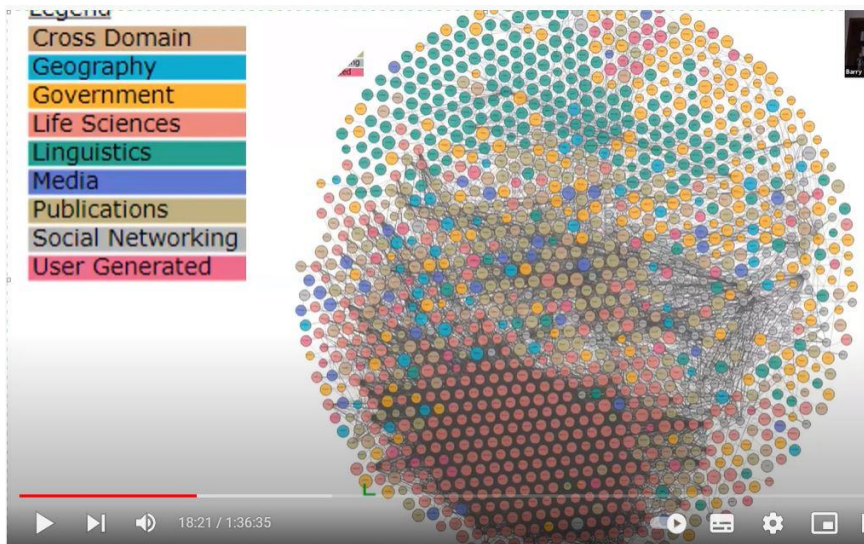


ICS > 01 > 01.040 > 01.040.35

ISO/IEC DIS 21838-2

INFORMATION TECHNOLOGY -- TOP-LEVEL ONTOLOGIES
(TLO) -- PART 2: BASIC FORMAL ONTOLOGY (BFO)

iso.org/standard/74572.html



Space and time

Actuality and possibility

Classes and types

Time and change

Parts, wholes, unity and boundaries

Space and place

Scale and granularity

Qualities and other attributes

Quantities and mathematical entities

Processes and events

Constitution

Causality

Information and reference

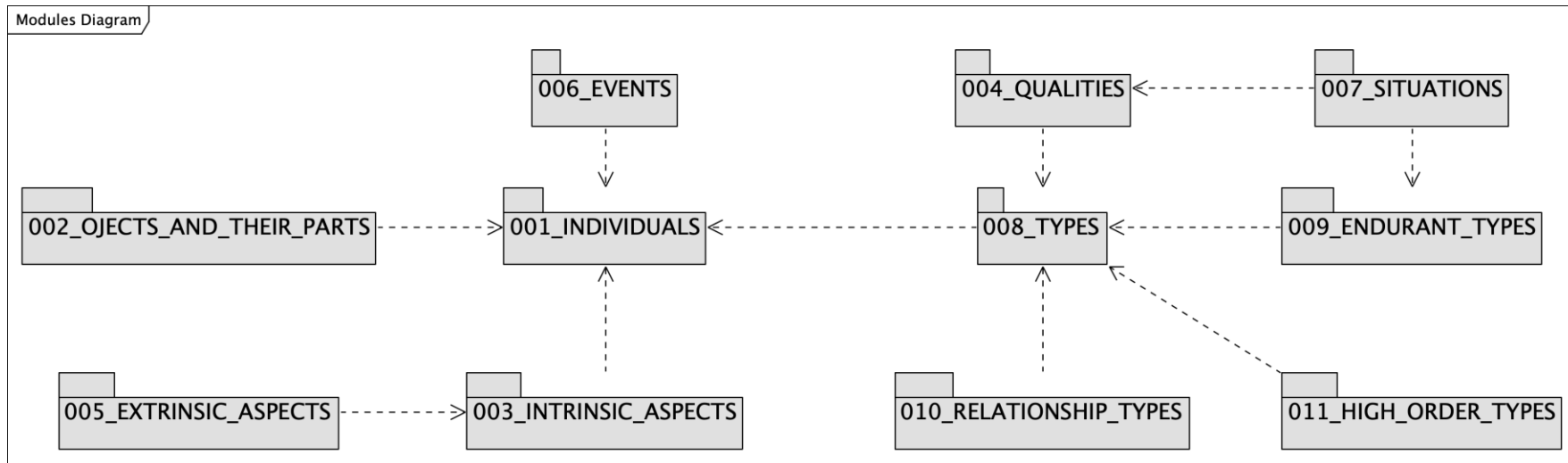
Artefacts and socially constructed entities

Mental entities; imagined entities; fiction; mythology; religion



CD ISO/IEC 21838-5 - Unified Foundational Ontology

<https://www.iso.org/standard/89915.html>



Suggest redrawing Figure 1 to place "types" vertically above "individuals" (i.e. instances) in hierarchies for each entity (e.g. objects, events, qualities, etc) and in accord with ISO704.



TC12
Quantities & units

ISO/TC 37

ISO/TC 37, Language and terminology

Liaison report to ISO/TC12 Quantities & Units

- *No progress*

L R Pendrill

ISO TC12 past Chair

Quantities & units in e-health

ISO/TC 215

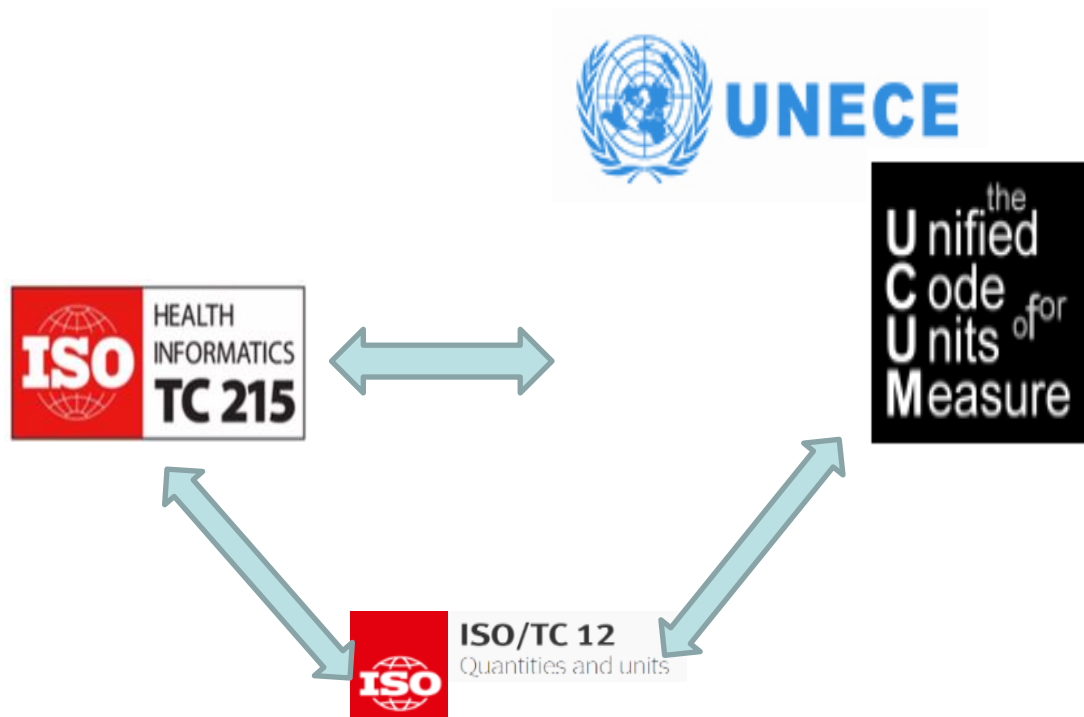


ISO/TC 215, ISO/TC 12 and IEC/TC 25

Quantities and Units in e-Health

GLOBAL SURVEY OF STANDARDS FOR
QUANTITIES AND UNITS USED/NEEDED IN
e-HEALTH

*Summary of Results, Recommendations
and Next Steps*





Quantities & units in informatics

CalConnect's TC UNITS manages the UnitsML standard used for encoding and representing units and quantities information in XML.



Originally developed at NIST and OASIS, UnitsML has now its formal home in CalConnect. Since the establishment of TC UNITS, UnitsML has seen several important adoptions, including within the OCX ship-building schemas as the authoritative method to encode units and quantities. Most recently, the BIPM's machine-readable SI Brochure has adopted UnitsML as its encoding and relies on the UnitsML to MathML rendering for presentation. The data set of units and quantities backing UnitsML, called UnitsDB, provides one of the most complete sets of units today.

In addition, TC UNITS is also interested in developing a machine-readable version of the units currently embedded inside the ISO and IEC 80000 publications.



ISO/TC 12
Quantities and units

CALCONNECT

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